

Pillar 5: Capstone Project

Learning Outcome Addressed

- Develop a robust foundational understanding of artificial intelligence (AI) and machine learning (ML) concepts and technologies.
- Equip learners with the knowledge and skills to identify and implement AI solutions across various sectors effectively.
- Develop proficiency in AI/ML tools and frameworks.
- Develop a holistic understanding of AI concepts and techniques, enabling them to effectively address complex real-world problems by designing, implementing, and evaluating AI and ML models.
- Develop hands-on skills in modelling, training, and deploying these models in real-world applications.

Assignment instructions:

- You must attempt all the given tasks.
 - This assignment carries a maximum of **100 points**.
 - Ensure clarity, depth, and relevance in your answers to maximize your score.
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To demonstrate the end-to-end application of the machine learning lifecycle—including problem framing, data preprocessing, modelling, evaluation, and result communication—on a real-world, industry-relevant dataset of choice.

Project Domains (Choose One)

Learners may choose from the following or propose a custom domain (with approval):

- **Healthcare:** Predict patient readmission or disease likelihood
 - **Finance:** Detect fraudulent transactions
 - **eCommerce:** Recommend products to users
 - **Education:** Predict student dropout risk
 - **Cybersecurity:** Detect anomalies in network traffic
 - **Clustering Option (Unsupervised):** Group customers, behaviors, or products based on similarity (K-Means, DBSCAN, Hierarchical)
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Step-by-Step Instructions

Step 1: Problem Understanding & Framing

- Frame the business and data science problem clearly.
- Define whether it's a classification, regression, recommendation, anomaly detection, or clustering task.
- Specify success metrics (e.g., Accuracy, AUC, RMSE, Silhouette Score) and business KPIs (e.g., cost savings, uplift).
- **Capstone linkage:** Module 1 output maps to Capstone Steps 1–3.

Deliverable: Clear problem statement + task type + target metric

Step 2: Data Collection & Understanding

- Use public datasets (Kaggle, UCI, APIs, etc.) or approved custom data.
- Summarize feature types, missing values, outliers, etc.
- Provide a **data dictionary** (variables, types, units, allowed values).

Deliverable: Dataset overview + data dictionary.

Step 3: Data Preprocessing, Applied EDA & Feature Engineering

- **Clean data:** Handle nulls, duplicates, and outliers.
- **Engineer features:** scaling, encoding, binning, and domain-derived features.
- **Applied EDA:** Distributions, relationships, clustering tendency (if unsupervised).
- **Feature importance & explainability:** SHAP, LIME, or model-based importances.
- **Feature selection:** At least one approach (filter, wrapper, or embedded).
- **Dimensionality reduction:** PCA (and t-SNE/UMAP for visualization if needed).

Deliverable: "EDA + Feature Engineering Report" with reproducible code & justifications.

Step 4: Model Implementation

- Experiment with appropriate models:
 - **Supervised:** Logistic Regression, Decision Trees, Random Forest, XGBoost, SVM, etc.
 - **Unsupervised:** K-Means, DBSCAN, Hierarchical (Elbow, Silhouette).
 - **Recommendation:** Collaborative or content-based.
 - **Deep Learning:** RNNs, CNNs, LSTMs, Transformers (if appropriate).
- Evaluation: compare with relevant metrics.
- **Reproducibility:** save configs and artifacts (models).

Deliverables: Trained models, metrics, and comparison between models.

Step 5: Critical Thinking → Ethical AI & Bias Auditing

- Explain model decisions (SHAP, LIME, PDP, ICE).
- Address limitations (imbalance, leakage, overfitting).
- **Bias detection & fairness audits:**
 - Check outputs across sensitive groups (gender, race, age, socioeconomic status).
 - Use fairness metrics (demographic parity, equalized odds, disparate impact).
 - Propose mitigations (reweighting, thresholds, augmentation, post-processing).

Deliverable: "Bias & Fairness Analysis" section in the final report.

Step 6: Final Presentation & Communication

- **Two deliverables for mixed audiences:**
 1. **Technical presentation** (Jupyter slides / LaTeX Beamer) → peers.
 2. **Business-facing presentation** (PowerPoint / Canva) → executives (ROI, risks, strategy).
 3. 8–12 slides per deck recommended.

Deliverables: Two slide decks (technical + business).

Step 7: GitHub Profile & Upload

- Create a public GitHub repo structured like an open-source project:
- Include: src/ for scripts, notebooks/, data/, models/ directories.

Deliverables: GitHub repo link + final report + reproducible code.

(Optional) Step 8: Deployment & MLOps

Complete this step if you have a good understanding of model deployment and MLOps practices.

- **Local deployment (required):** deploy the best model via **Flask**, **FastAPI**, or **Dash**.
- **Optional cloud:** AWS SageMaker, GCP Vertex AI, or Azure ML.
- **MLOps practices:**
 - Reproducible environments (requirements.txt, Docker).
 - Config-driven runs & experiment tracking (MLflow, W&B, etc.).
 - CI checks (lint, unit tests), basic monitoring plan.
 - Versioning & rollback plan.
- Provide a **demo** (GIF/screencast).

Deliverables: Running app + deployment guide + demo media.

(Optional) Step 9: Use of Generative AI

You may optionally use Generative AI tools to:

- Use LLMs to auto-generate EDA summaries or data dictionaries.
- Build GenAI-enhanced applications (e.g., LLM-backed recommenders, chatbots)

Deliverable:

- Document how Generative AI was used in the project
- Include code + examples in the GitHub repo or presentation
- Demo video.

Bonus points (up to +5) will be awarded for a creative and well-presented submission.

Submission Instructions:

- Go through the instructions and evaluation rubric to understand what is expected from this assignment.
- We recommend collating all the textual responses in a file. Record your responses in an approved format: .pdf, .doc, .pptx, or .ppt.
- You are expected to submit the coding files.
- Rename the files as **Your_Name_Assignment name**.
- After you complete your assignment, select the **Start Assignment** button at the top of this page.
- Upload the file containing your responses.
- Select the **Submit Assignment** button to submit your response.

Pillar 5: Capstone Project: Rubric		
Criteria	Ratings	Pts
1: Problem Understanding & Framing	10 to >5.0 pts Outstanding/Exemplary • Problem clearly framed with strong business context and data science perspective. • Task type (classification/regression/etc) correctly identified and justified. • Success metrics (technical + business KPIs) are relevant, measurable, and well-explained.	10 pts
Step 2: Data Collection & Understanding	10 to >5.0 pts Outstanding/Exemplary • High-quality dataset chosen and justified (source cited). • Comprehensive dataset overview: feature types, missing values, outliers, distributions. • Clear, complete data dictionary (variables, types, ranges/units).	10 pts
Step 3: Data Preprocessing, EDA & Feature Engineering	10 to >5.0 pts Outstanding/Exemplary • All preprocessing steps documented with reproducible code. • Clear handling of nulls, outliers, and duplicates. • Insightful applied EDA with visuals, distributions, and correlations. • Feature engineering shows domain knowledge and creativity. • At least one feature selection + dimensionality reduction method used and justified.	10 pts
Step 4: Model Implementation & Comparison	20 to >10.0 pts Outstanding/Exemplary • Multiple models implemented and tuned appropriately. • Evaluation metrics correctly applied and compared across models. • Reproducibility ensured (saved models/configs). • Clear reasoning for model choice based on results.	20 pts
Step 5: Critical Thinking, Ethical AI & Bias Auditing	20 to >10.0 pts Outstanding/Exemplary • Excellent use of explainability tools (SHAP/LIME/PDP/ICE). • Thorough discussion of data/model limitations (imbalance, leakage, overfitting). • Bias audit performed across sensitive groups with fairness metrics. • Proposes clear, feasible mitigation strategies.	20 pts
Step 6: Final Presentation & Communication	10 to >5.0 pts Outstanding/Exemplary • Two high-quality, well-structured presentations (technical + business). • Technical deck: clear methodology, visuals, metrics. • Business deck: ROI, risks, strategy clearly communicated for a non-technical audience. • Visually professional and concise (8–12 slides per deck).	10 pts
Step 7: GitHub Profile & Upload	15 to >10.0 pts Outstanding/Exemplary • Public GitHub repo structured like an open-source project (src/, notebooks/, data/, models/). • Includes README, requirements.txt, final report, reproducible code. • Clean, professional commit history.	15 pts
Bonus points: Creative and well-presented submission	5 to >2.5 pts Outstanding/Exemplary Demonstrates exceptional creativity, originality, and clear presentation. The submission goes beyond expectations in terms of design, clarity, or innovation.	5 pts
		Total Points: 100